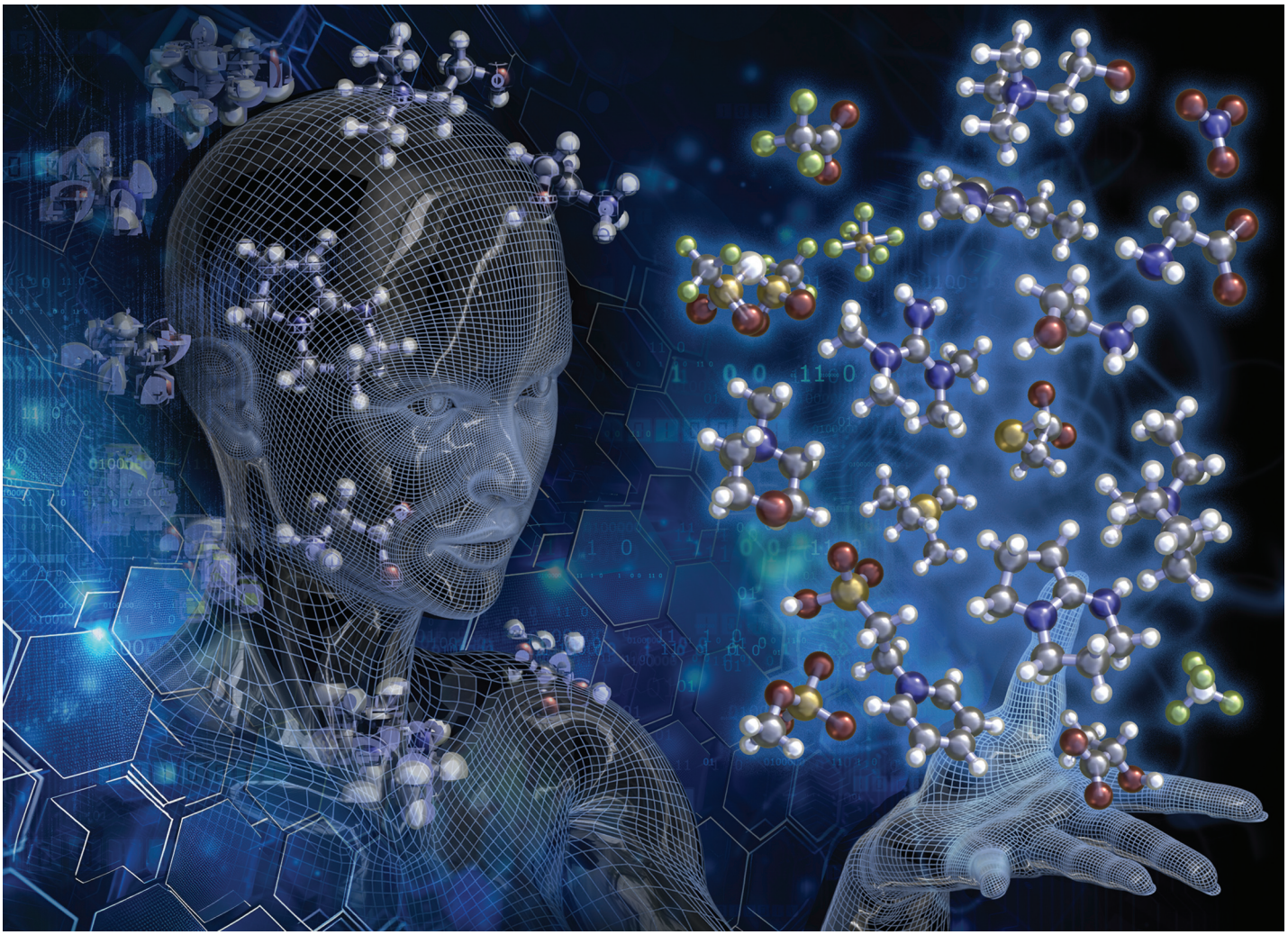

Showcasing research from the Biofuels Science Focus Area team, Oak Ridge National Laboratory, Tennessee, USA, and Prof. Jeremy C. Smith, Center for Molecular Biophysics, Oak Ridge National Laboratory, TN, USA.

Molecular property prediction for very large databases with natural language processing: a case study in ionic liquid design

Machine learning with natural language processing–based molecular embeddings is used to represent ionic liquid structures, enabling accurate prediction of key physicochemical properties. This approach allows rapid screening of more than ten million ionic liquids to identify task-specific candidates for applications including biomass processing, CO_2 capture, and battery electrolytes, accelerating the design of efficient and sustainable solvents.

Cover artwork created by Andrew Sproles.

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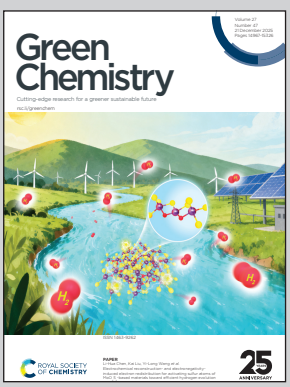

See Mood Mohan, Jeremy C. Smith *et al.*, *Green Chem.*, 2025, **27**, 15106.

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